Do Now · retrieval

Lesson 099 · CISC and RISC: Instruction Set Design

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is ISA?

Q2

What does an opcode identify?

Q3

What is a load/store instruction?

Q4

Simplify $(A \wedge B) \vee (A \wedge \neg B)$.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — CISC typically offers a larger/more complex instruction repertoire, with instructions able to combine operations; RISC generally uses more regular, often simple operations.

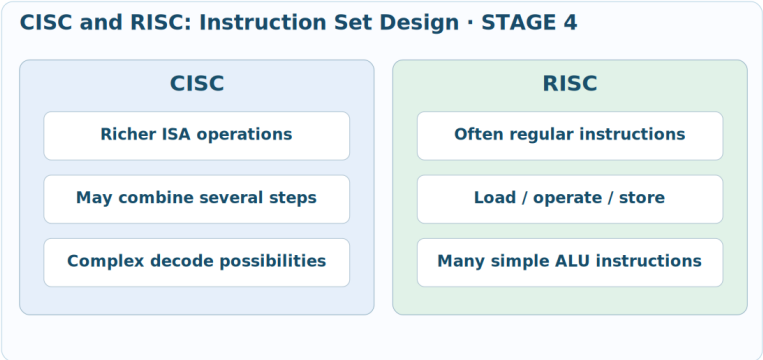
Core idea 2

Apply — RISC load/store designs explicitly move operands between memory and registers and perform ALU work using registers.

Core idea 3

Evaluate — Avoid outdated absolutes: modern CISC and RISC microarchitectures can both pipeline, use caches and achieve strong performance; trade-offs depend on implementation.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ISA — Programmer-visible machine instruction set.

CISC — Complex Instruction Set Computer.

RISC — Reduced Instruction Set Computer.

load/store — Explicit memory-register data transfer.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

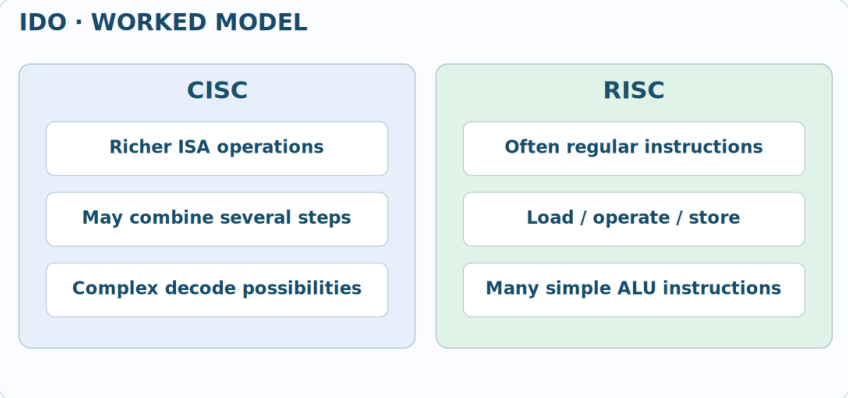
Problem

Model a RISC load-add-store sequence for A5,B8 and result C.

Follow the worked method

1. Follow the reasoning — LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
R1=4,R2=7,R3=11, memory[C]=11.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State two common RISC and two common CISC characteristics. [4]

Question 2

Trace the abstract RISC sequence above when A=6,B=9, give R1,R2,R3,C. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate statement 'CISC processors cannot use pipelining and RISC always uses fewer instructions'. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Modern CISC implementations cannot use pipeline stages.

Use key vocabulary from memory

1. Explain ISA in your own words.

2. Explain CISC in your own words.

3. Explain RISC in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.